Chenyang LIU

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Education

Duke University — Durham, USA
PhD in Computer Science, advised by Prof. Matthew Lentz and Prof. Kartik Nayak — *Aug. 2024 – Now*

Shanghai Jiao Tong University — Shanghai, China
B.Sc. GPA: 90.33/100 Rank: 3/86 — *Sep. 2020 – June 2024*
Member of Zhiyuan Honors Program of Engineering in Zhiyuan College; Top 5%

Research Interest

Consensus protocols, DeFi, Trust Execution Environments (TEEs).

Research Experiences

Duke University — Durham, USA
Initial Project advised by Prof. Matthew Lentz and Prof. Kartik Nayak — *Aug. 2024 – Present*

Decentralized Proposer-as-a-Service

- **Removed Relays in Ethereum PBS**: Removed centralized relays and outsourced the combined role of proposer and relays to a group of TEEs (PEs)
- **Designed a new primitive Crab-BB**: a generalized Byzantine Broadcast protocol achieving both censorship-resistance and backed-availability for any application semantic including Ethereum PBS.
- **Implemented DPaaS**: validate the functionality in a test-net and evaluate the prototype, showing 3% more MEV-earning in DPaaS than relays.

Shanghai Jiao Tong University — Shanghai, China
Undergraduate Research Assistant advised by Prof. Guoxing Chen — *Feb. 2023 – Apr. 2025*

Closing Cache Side Channels via Mimimal Cache Fingerprints

- **Memory Footprint Minimization**: Designed a software-simulated and fully-associated cache architecture to hide access pattern, enhancing resistance to access-pattern based leakages
- **Compiler Modification**: Instrumented enclave code into 64-bytes blocks by modifying LLVM backend
- **Verification of the OS**: Proposed a novel contract to verify the OS under SGX's threat model

Publications

DPaaS: Improving Decentralization by Removing Relays in Ethereum PBS — In Submission
*<u>**C Liu**</u>, I Abraham, M Lentz, K, Nayak*

A Formal Approach to Multi-Layered Privileges for Enclaves — NDSS 2025
*G Yang, <u>**C Liu**</u>, Z Huang, G Chen, H Fu, Y Zhang, H Zhu*

Teaching Services

Teaching Assistant — Duke University
Undergraduate Course: Introduction to Operating System — *Spring 2025*

Teaching Assistant — Duke University
Graduate Course: Operating System — *Fall 2025*

Selected Honors & Awards

Zhiyuan Honor Awards: Awards to students in Zhiyuan Honor Program (Top 5%) — 2021 – 2023
Academic Progress Awards: Awards to students who have made great academic progress in one year (3/88) — 2022